Design Verification Plan and Report — t8_r3 (VBF-T8R3-DVPR-01)

Design Verification Plan and Report (virtual test version) — t8_r3 (VBF-T8R3-DVPR-01)

All results below are simulation outputs (no physical cell exists); uncovered conditions marked N/A honestly.

	Condition	Result	Determination	Sou
	1C discharge energy ÷ stack mass	497.54 Wh/kg	✓ pass vs ≥446.18	ene
	5C DFN capacity ÷ 1C DFN capacity	98.66%	✓ pass vs ≥90%	deri
	calc-energy stack mass	37.32 g	✓ pass vs ≤40 g	ene
e rise	4C charge, 45 °C chamber, thermal lumped	326.53 K	✓ pass vs ≤333.15 K	4c_
	same run, anode_potential_v vs 0 V	min +0.0177 V → plated=false	✓ pass	4c_
	1C CC discharge to 2.5 V	5.0728 Ah	informational (no capacity criterion in entry 0)	1c_
	parameter set limits	2.5–4.2 V	within parameter-set bounds	Che
away	overcharge + 3-reaction ODE, T0=overcharge T_max	triggered=False (299.33 K → 299.33 K)	✓ not triggered (informational)	ove
	—	N/A (beyond pure simulation boundary, requires physical experiment)	N/A	—
	—	N/A (requires physical experiment)	N/A	—
	aging protocol not in objective	N/A (not simulated — no entry-0 criterion)	N/A	—

Conclusion: all five entry-0 criteria pass (see rows 1–5); safety-related informational checks pass; nail/crush/drop/cycle-life remain beyond virtual-testing boundary.